all of that comes under the Data Analytics specialisation.

Networks and Architecture – An IoT system can be viewed as a complicated mesh of connected gadgets and objects that ultimately makes no sense if it is not planned properly before implementation. Due to the wide variety of implementations being done and will be possible in the future, there are different types of sensors and transmitters that communicate differently within the system. This is where the network specialisation would come in. There will be a broad array of methods of communicating data. Networking specialists have been dealing with computer networks so far, and compared to IoT networks, that's a piece of cake. Going forward, they will need to deal with large scale traffic across secure, reliable and redundant backbones between connected locations. This is the very reason why there is a dire need for networking specialists who can not only contribute to implementing the IoT network, but can take part in its design as well. Even now, completely new networking protocols are being designed to work optimally with low power sensors and devices. Designing the architecture of the IoT network is a crucial aspect of any IoT project and investing your time into developing an expertise in this segment might pay off big time. An IoT architect would be a very important person in any IoT implementation. The guy who understands the systems and how they will interact, who understand the devices industry from an overview, that person shall be a key component of the process.

Security – This is the current buzzword within IoT. With the sudden explosion of device and sensor implementation, the industry has only now realised that all that data and all those devices also need to be protected from malicious external sources. Not like it's already a very matured sector in terms of experts available. And with the number of IP-enabled devices growing by the day, cyber-security specialists will be even more sought after in the job market. If the security implementation on your smart fridge is weak, and it is connected to the same network as your laptop, it might be quite possible, and in fact, easy for a hacker to use this path to your confidential data. Some of the key skills in this area are risk identification, vulnerability analysis, Public Key Encryption security and wireless network security.

Device and Hardware – Hardware engineers are the people who actually put together the various components available to manufacture the device in terms of design. The same is applicable to IoT as well, albeit with a large number of sensors and transmitters. Not only will the conventional manufacturing of these sensors increase, there is already a high demand to design newer sensors and components that run on lower power with higher efficiency. Also, engineers and device experts who can implement competent Wi-Fi, Bluetooth and other connectivity solutions are also highly sought after. The designers will also need to be strong with their soft skills, as they will have to work closely with the manufacturing and planning teams.

Mobile and UI development – The IoT boom has come at a time where our lives are closely integrated with smartphones. And since the whole point of IoT is to connect everything all-the-time, smartphones and mobile devices are ideal candidates for the platform of choice to control IoT devices. Needless to say, this means there is a high demand for Android and iOS developers in IoT. Not that those two roles actually needed any additional demand, but current developers will need to gain an expertise in working with programming libraries that allow apps to communicate with external devices and sensors. This demand also extends to User Interface and User Experience designers, with devices and screens of all shapes and sizes showing up in the IoT world. Providing a responsive and persistent user experience will soon be an even more sought after skill than it is currently.

MR. RANJIT NAIR
CO-FOUNDER AND CTO
ALTIZON SYSTEMS

"IoT requires you to look at hardware from the viewpoint of security, inter-connectivity and interoperability with software systems"

According to Mr. Ranjit Nair, Co-Founder and CTO, Altizon Systems, all the above segments need to cohesively work together and are equally important in terms of career selection. In his words, "IoT requires you to look at hardware from the viewpoint of security, inter-connectivity and interoperability with software systems. This means that hardware will get redesigned using chipsets that are more IoT friendly. Usage of connectivity protocols ranging from RF, Wi-FI, Zigbee, BLE to classical wired protocols will see an explosion as intelligent devices start talking back. Processing and analysing this data will require new analytical models as there aren't pre-existing models to fall back on. IoT will place a huge demand on each of these areas. IoT hardware will end up being a commodity sooner rather than later but software and analytics will evolve for a longer duration".

Why IoT in India matters right now?

Multiple surveys from leading global firms like Gartner, KPMG and device manufacturers like Cisco have predicted anything between 25 billion to 50 billion connected devices by 2020. Considering the significance of this number, the Government of India has given a lot of importance to IoT and it's growth in India in its Digital India initiative. The government's vision comprises of building an IoT industry worth $15 billion by the end of this decade, and for that they have set out a strategic plan to increase domain proficiency, promote